

Regular/Supplementary Winter Examination – 2024

Subject Code & Name: Software Engineering (BTECPE503D)

Max Marks: 60

Date:11/02/2025

Duration: 3 Hr.

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

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5	What is a Functional Requirement?	L-1/CO2	1
	a. Specifies the tasks the program must complete b. Specifies the tasks the program should not complete c. Specifies the tasks the program must not work d. All of the mentioned		
6	Which is not a step of Requirement Engineering?	L-1/CO2	1
	a. Requirements elicitation b. Requirements analysis c. Requirements design d. Requirements documentation		
7	The _____ model helps in representing the system's dynamic behavior.	L-1/CO3	1
	a. Object Model b. Context Model c. Behavioral Model d. Data Model		
8	Which design identifies the software as a system with many components interacting with each other?	L-1/CO3	1
	a. Architectural design b. High-level design c. Detailed design d. Both B & C		
9	If the objects focus on the problem domain, then we are concerned with _____.	L-1/CO4	1
	a. Object Oriented Analysis b. Object Oriented Design c. Object Oriented Analysis and Design d. None of the above		
10	Which of the following is not open source software?	L-1/CO4	1
	a. LibreOffice b. Microsoft Office c. GNU image manipulation d. MySQL		

11	Software testing is:-----	L-1/CO5	1
	a. the process of demonstrating that errors are not present		
	b. the process of establishing confidence that a program does what it is supposed to do		
	c. the process of executing a program to show it is working as per specifications		
	d. the process of executing a program with the intent of finding errors		
12	Which of the following testing is the part of non-functional testing?	L-1/CO5	1
	a. Unit Testing		
	b. Performance Testing		
	c. System Testing		
	d. Integration Testing		
Q. 2	Solve the following.		12
A)	Describe incremental approach for software development.	L-2/CO1	6
B)	Explain in brief characteristics of software.	L-3/CO1	6
Q.3	Solve the following.		12
A)	Define Agility. List any five principles of Agility.	L-2/CO2	6
B)	Explain and compare functional and nonfunctional requirements in Software Engineering.	L-3/CO2	6
Q. 4	Solve Any Two of the following.		12
A)	What do you mean by system modeling? Explain Context Model.	L-3/CO3	6
B)	Discuss the layered architecture pattern with four layers.	L-2/CO3	6
C)	Explain Pipe and filter architecture.	L-3/CO3	6
Q.5	Solve Any Two of the following.		12
A)	Explain the process of Object-oriented design using UML.	L-3/CO4	6
B)	Describe essential elements of Design patterns.	L-2/CO4	6
C)	Write a short note on open source development.	L-1/CO4	6

Q. 6	Solve Any Two of the following.		12
A)	Define software testing. Which are the different benefits of software testing?	L-2/CO5	6
B)	What is the difference between validation and verification?	L-2/CO5	6
C)	Explains safety requirements and safety cases in software testing.	L-3/CO5	6
*** End ***			